ClipSAT : Cambridge & Edexcel Additional Mathematics 9606/4PM1 [ANSWER KEY]

Unit 5.01: Trigonometric Functions & Circular Measure

Q1. A sector of a circle of radius r = 8 text{ cm} has central angle theta = 0.9 text{ rad} . What is the area of the sector?

[x] A: 28.8 text{ cm}^2
[ ] B: 57.6 text{ cm}^2
[ ] C: 33.8 text{ cm}^2
[ ] D: 7.2 text{ cm}^2

Solution: **Mark Scheme CAIE / Edexcel Grading Steps:** - **Step 1 Method & Formulation:** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]** - **Step 2 Derivation & Calculation:** Sector area formula: A = frac{1}{2}r^2theta = frac{1}{2}8^20.9 = 28.8 text{ cm}^2 . **[M1]** - **Step 3 Final Accuracy:** Verified exact result: 28.8 text{ cm}^2 . **[A1]**

Q2. A sector of a circle of radius r = 10 text{ cm} has central angle theta = 1.0 text{ rad} . What is the area of the sector?

[x] A: 50.0 text{ cm}^2
[ ] B: 100.0 text{ cm}^2
[ ] C: 55.0 text{ cm}^2
[ ] D: 10.0 text{ cm}^2

Solution: Sector area formula: A = frac{1}{2}r^2theta = frac{1}{2}10^21.0 = 50.0 text{ cm}^2 . [M1][A1]

Q3. A sector of a circle of radius r = 12 text{ cm} has central angle theta = 1.1 text{ rad} . What is the area of the sector?

[x] A: 79.2 text{ cm}^2
[ ] B: 158.4 text{ cm}^2
[ ] C: 84.2 text{ cm}^2
[ ] D: 13.2 text{ cm}^2

Solution: Sector area formula: A = frac{1}{2}r^2theta = frac{1}{2}12^21.1 = 79.2 text{ cm}^2 . [M1][A1]

Q4. A sector of a circle of radius r = 14 text{ cm} has central angle theta = 1.2 text{ rad} . What is the area of the sector?

[ ] A: 16.8 text{ cm}^2
[x] B: 117.6 text{ cm}^2
[ ] C: 235.2 text{ cm}^2
[ ] D: 122.6 text{ cm}^2

Solution: Sector area formula: A = frac{1}{2}r^2theta = frac{1}{2}14^21.2 = 117.6 text{ cm}^2 . [M1][A1]

Q5. A sector of a circle of radius r = 16 text{ cm} has central angle theta = 1.3 text{ rad} . What is the area of the sector?

[ ] A: 332.8 text{ cm}^2
[ ] B: 20.8 text{ cm}^2
[x] C: 166.4 text{ cm}^2
[ ] D: 171.4 text{ cm}^2

Solution: Sector area formula: A = frac{1}{2}r^2theta = frac{1}{2}16^21.3 = 166.4 text{ cm}^2 . [M1][A1]

Unit 5.01: Trigonometric Functions & Circular Measure

Q6. A sector of a circle of radius r = 18 text{ cm} has central angle theta = 1.4 text{ rad} . What is the area of the sector?

[ ] A: 453.6 text{ cm}^2
[ ] B: 25.2 text{ cm}^2
[ ] C: 231.8 text{ cm}^2
[x] D: 226.8 text{ cm}^2

Solution: Sector area formula: A = frac{1}{2}r^2theta = frac{1}{2}18^21.4 = 226.8 text{ cm}^2 . [M1][A1]

Q7. A sector of a circle of radius r = 20 text{ cm} has central angle theta = 1.5 text{ rad} . What is the area of the sector?

[x] A: 300.0 text{ cm}^2
[ ] B: 600.0 text{ cm}^2
[ ] C: 305.0 text{ cm}^2
[ ] D: 30.0 text{ cm}^2

Solution: Sector area formula: A = frac{1}{2}r^2theta = frac{1}{2}20^21.5 = 300.0 text{ cm}^2 . [M1][A1]

Q8. Find all solutions of 2cos^2 x + 3sin x - 3 = 0 in the interval 0 le x le 2pi .

[x] A: x = frac{pi}{6}, ; frac{5pi}{6}, ; f
[ ] B: x = frac{pi}{3}, ; frac{2pi}{3}
[ ] C: x = frac{pi}{6}, ; frac{7pi}{6}
[ ] D: x = 0, ; pi

Solution: Solving quadratic trig equation 2cos^2 x + 3sin x - 3 = 0 yields solutions x = frac{pi}{6}, ; frac{5pi}{6}, ; frac{pi}{2} . [M1][A1]